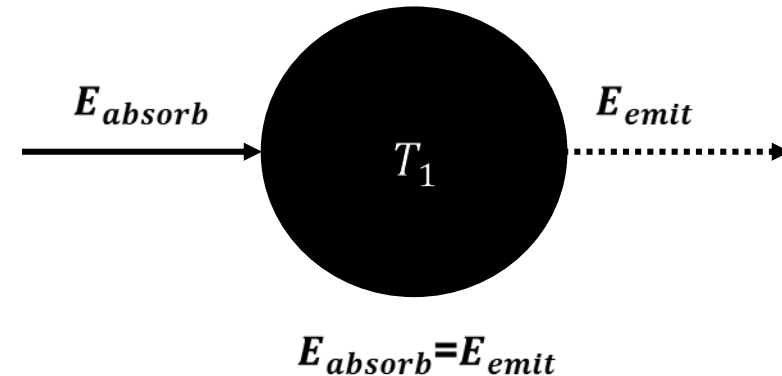


# Quantization of Energy

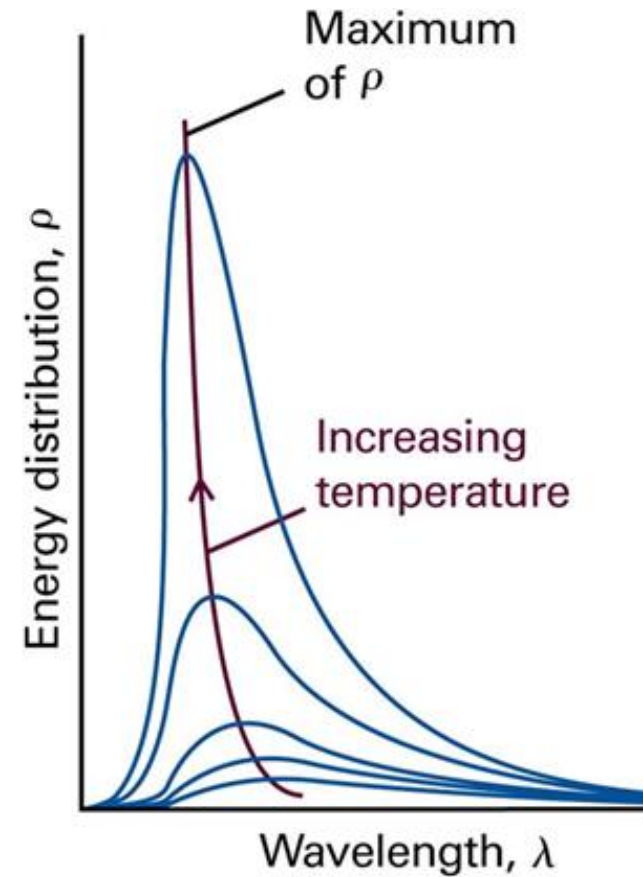
# Black Body Radiation

What is Black Body ?

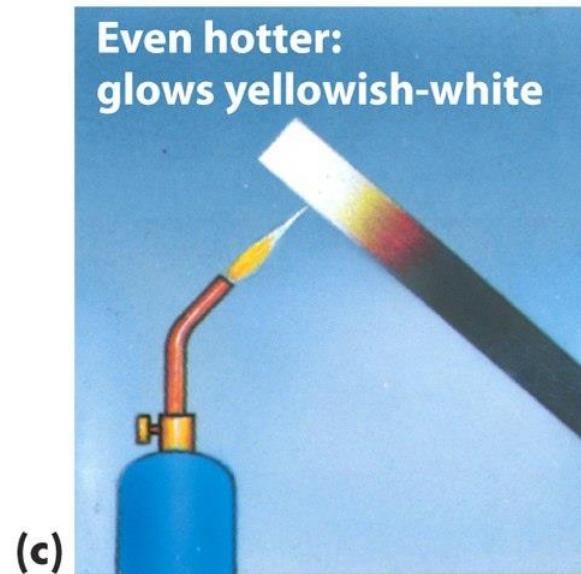
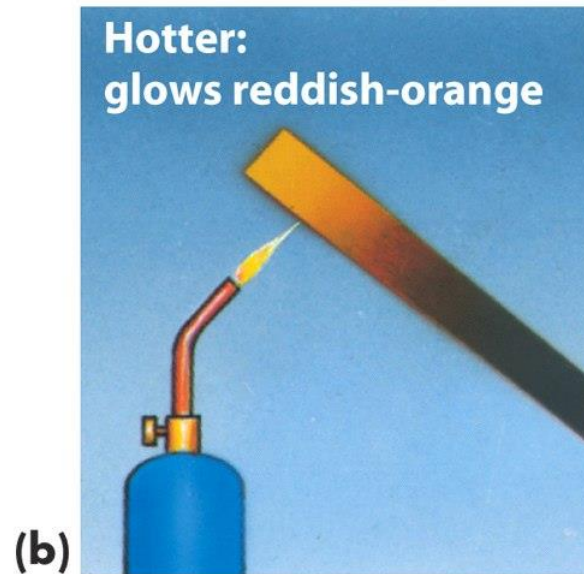
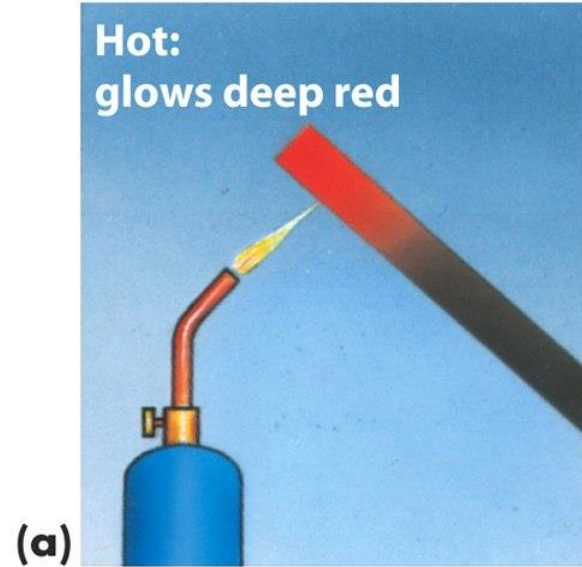
- A blackbody is a theoretical or model body which absorbs all radiation falling on it, reflecting or transmitting none.
- It is a hypothetical object which is a “perfect” absorber and a “perfect” emitter of radiation over all wavelengths.



- A hot object emits electromagnetic radiation of varying frequencies depending on the nature of the object and its temperature.
- As the temperature increases, the peak wavelength emitted by the black body decreases.
- As temperature increases, the total energy emitted increases



An opaque object emits electromagnetic radiation according to its temperature



# Planck Spectrum

As an object is heated, the radiation it emits peaks at higher and higher frequencies

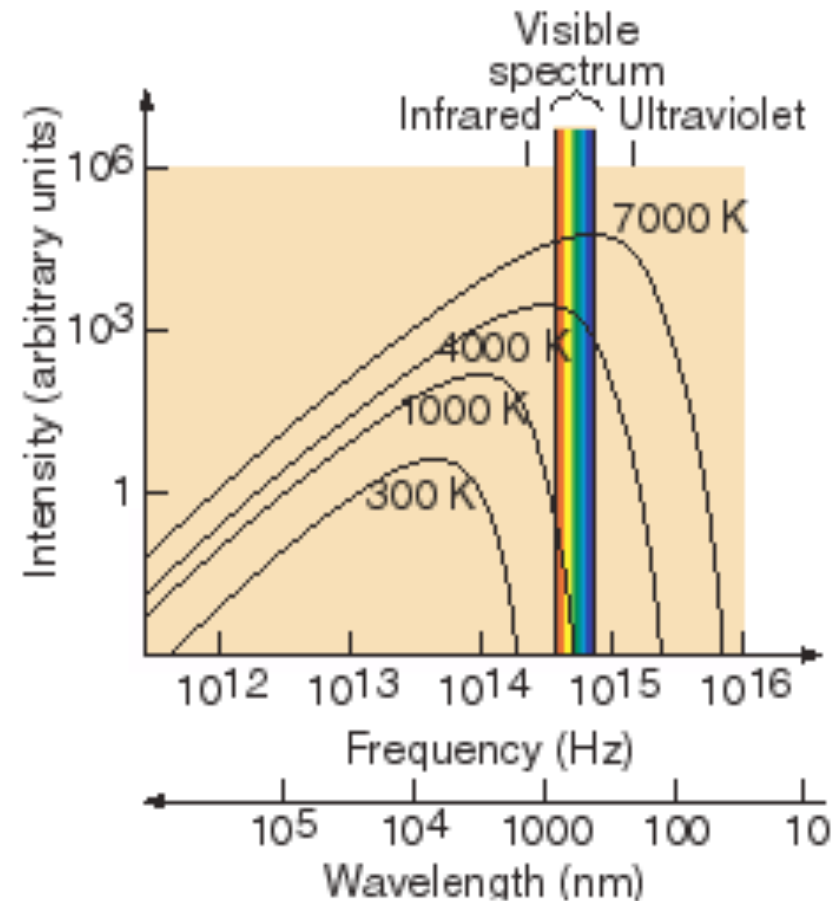
Shown here are curves corresponding to temperatures of

300 K (room temperature),

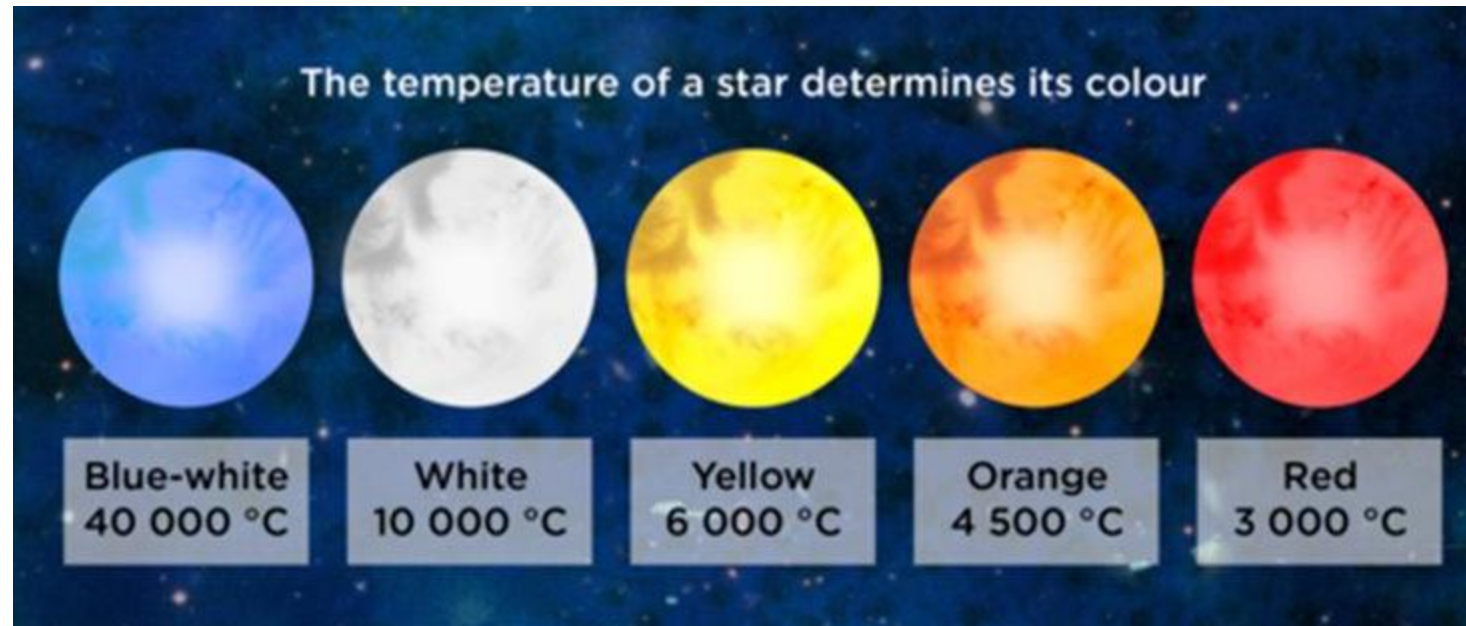
1000 K (begin to glow deep red)

4000 K (red hot)

7000 K (white hot).

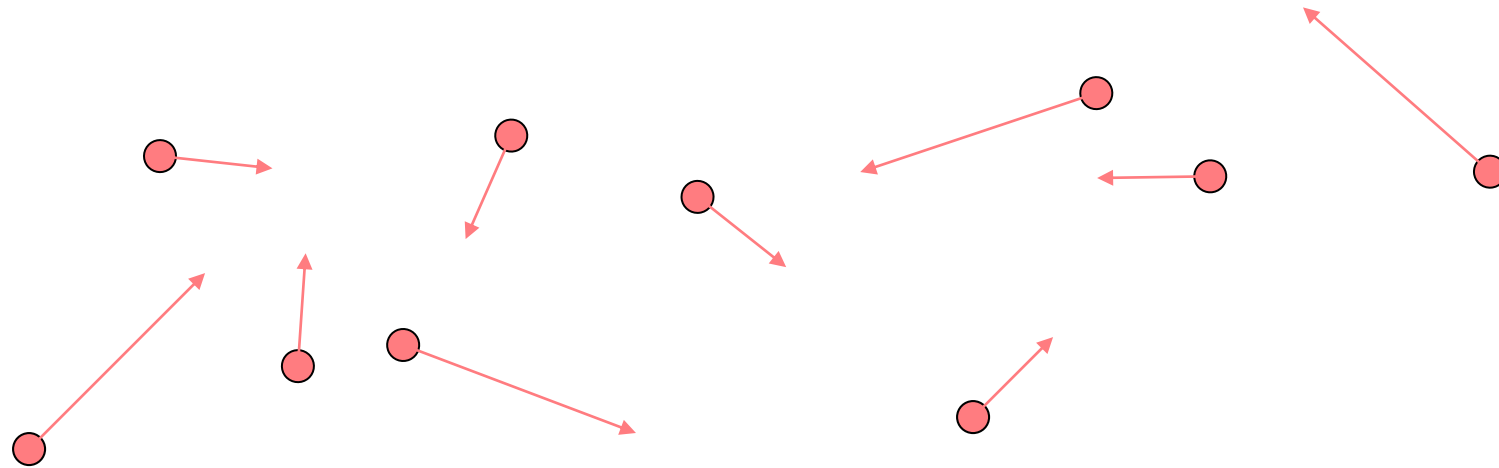


- Red stars are relatively cool. A yellow star, such as our own sun, is hotter. A blue star is very hot.



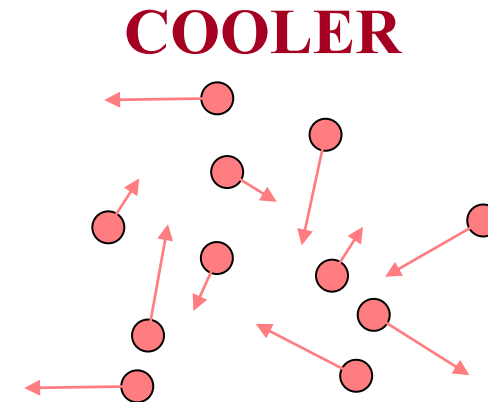
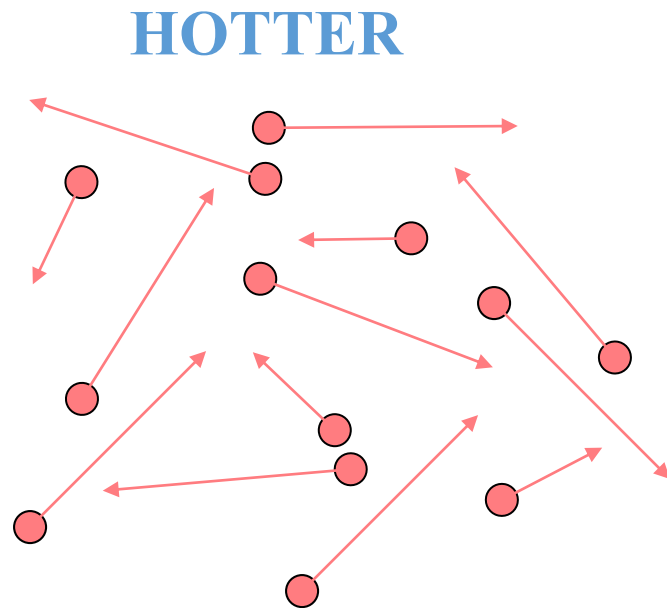
# Atoms in Motion

- Everything is composed of atoms which are constantly in motion.



# Temperature

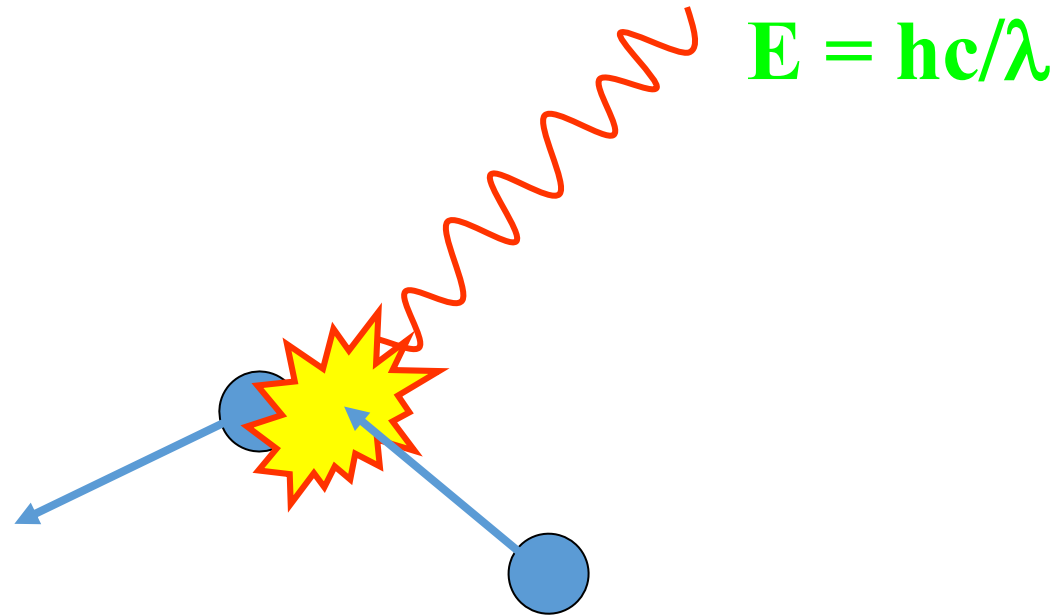
- The hotter the object, the faster the average motion of the atoms.





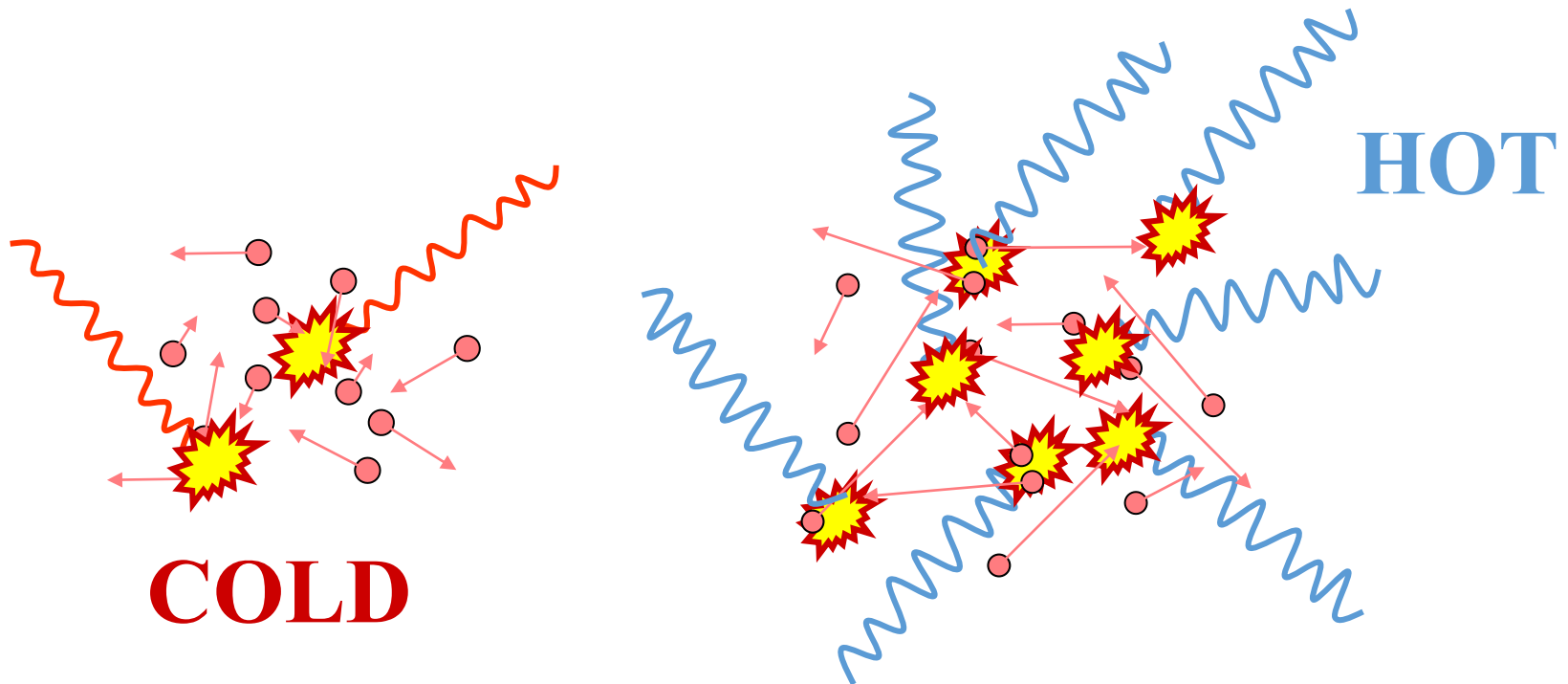
# Atoms and Light

- As atoms move they collide (interact, accelerate).
- Collisions give off energy.
- But light IS energy.



# Light and Temperature

- The hotter the object the faster the average atom and the more energetic the average collision.
- The faster the atoms the more collisions there are.



# Energy and Intensity

- The more energetic the average collision the bluer the average light that is given off.
- Since  $E = hc/\lambda$
- The more collisions that occur the more light that is given off per surface area.

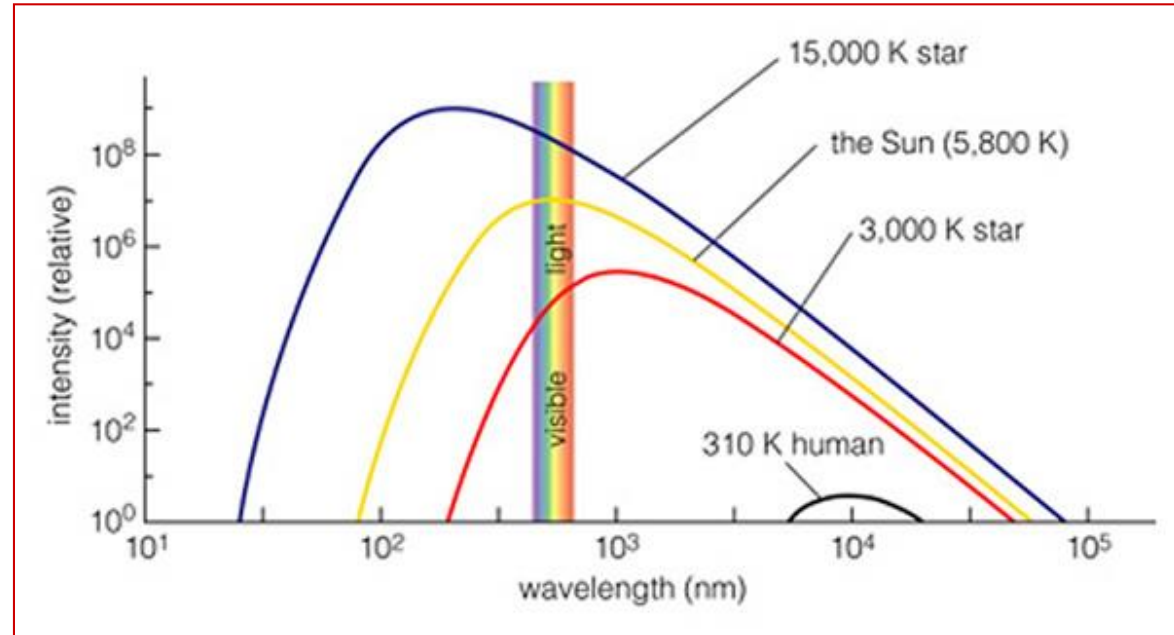
# Thermal Radiation Laws

## 1. Hotter is bluer.

- (peak at shorter wavelength)

## 2. Hotter is brighter.

- (More intense at all wavelengths)



- The characteristics of blackbody radiation can be described in terms of several laws:

2. Wien's Displacement Law

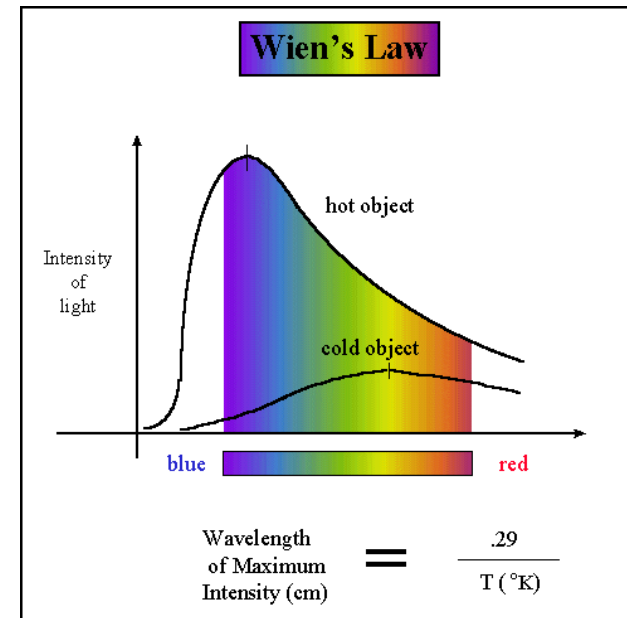
3. Stefan–Boltzmann Law

# WIEN'S DISPLACEMENT LAW

**Wien's Displacement Law** states that the frequency of the peak of the emission ( $f_{\max}$ ) increases linearly with absolute temperature (T).

As the temperature of the body increases, the wavelength at the emission peak decreases.

$$f_{\max} \propto T$$



# Stefan-Boltzmann Law

Stefan–Boltzmann Law, which relates the total energy emitted ( $E$ ) to the absolute temperature ( $T$ ).

$$E \propto T^4$$

- The blackbody radiation curves have quite a complex shape (described by Planck's Law).
- The spectral profile (or curve) at a specific temperature corresponds to a specific peak wavelength, and vice versa.
- As the temperature of the blackbody increases, the peak wavelength decreases (Wien's Law).

# Rayleigh-Jean's law

- The energy density  $p_\nu$  per unit frequency interval at a frequency  $\nu$  is, according to the The Rayleigh-Jeans Radiation,

$$\rho_\nu = \frac{8\pi\nu^2}{c^3} \langle E \rangle$$

$$\langle E \rangle = kT$$

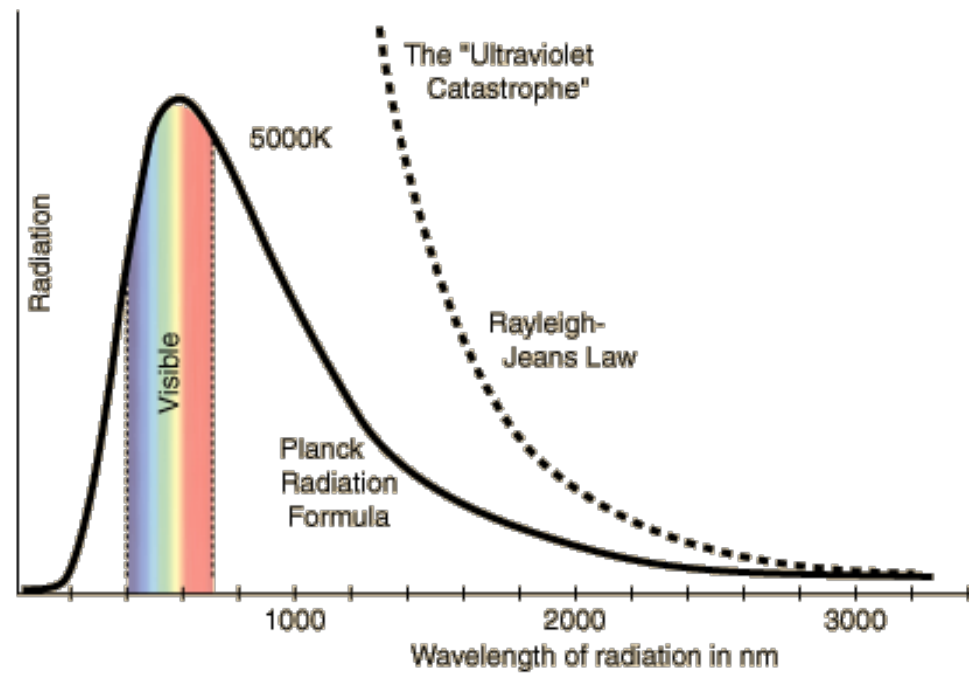
$$d\rho(\nu, T) = \rho_\nu(T) d\nu = \frac{8\pi kT}{c^3} \nu^2 d\nu$$

$$d\rho(\lambda, T) = \frac{8\pi kT}{c^3} \left(\frac{c}{\lambda}\right)^2 \left(\frac{cd\lambda}{\lambda^2}\right)$$

$$d\rho(\lambda, T) = \frac{8\pi kT}{\lambda^4} d\lambda$$



- Classical physics would predict that even relatively cool objects should radiate in the UV and visible regions. In fact, classical physics predicts that there would be no darkness



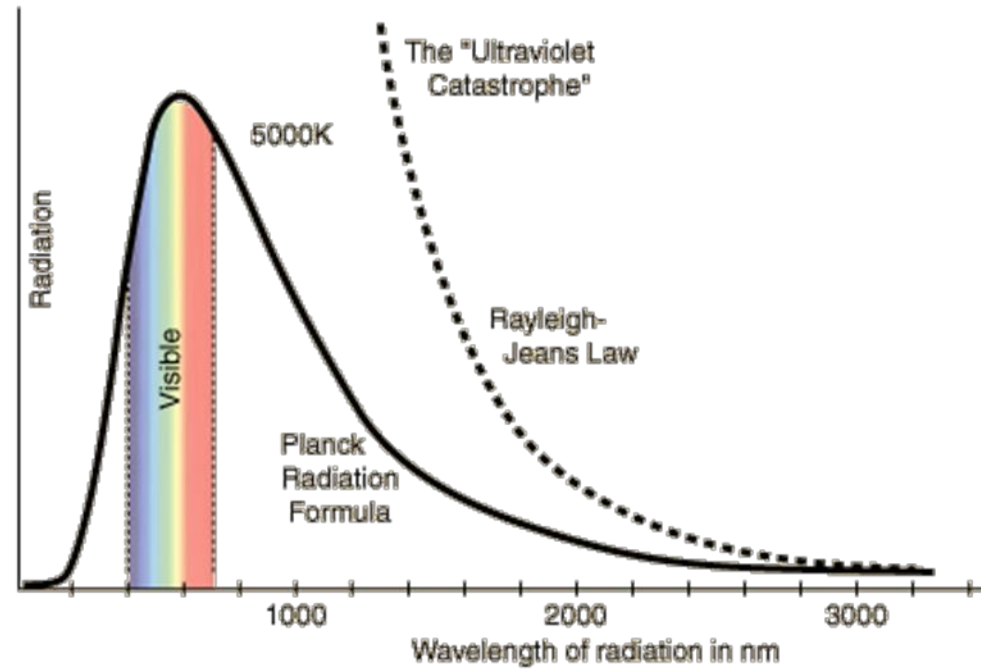
- Max Planck explained the blackbody radiation in 1900 by assuming that the energies of the oscillations of electrons which gave rise to the radiation must be proportional to integral multiples of the frequency, i.e.,  $E = nh\nu$
- The recognition that energy changes in discrete quanta at the atomic level marked the beginning of quantum mechanics
- Planck won the Nobel Prize in Physics in 1918

# Planck's Concept

- The average energy per "mode" or "quantum" is the energy of the quantum times the probability that it will be occupied

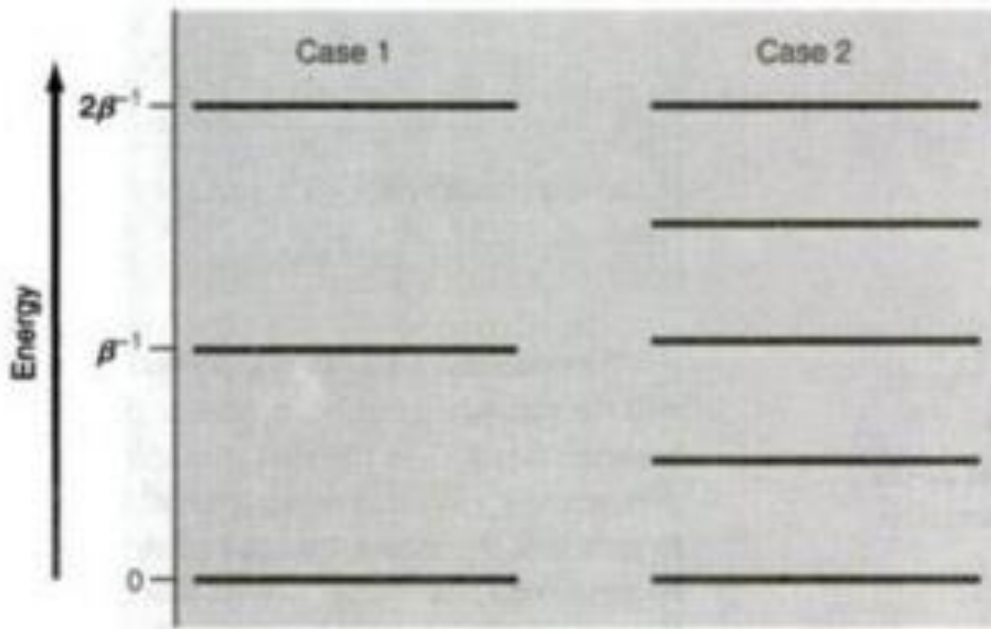
$$\langle E \rangle = \frac{h\nu}{e^{h\nu/kT} - 1}$$

$$d\rho(\nu, T) = \rho_\nu(T) d\nu = \frac{8\pi\langle E \rangle}{c^3} \nu^2 d\nu = \frac{8\pi h}{c^3} \frac{\nu^3}{\exp\left(\frac{h\nu}{kT}\right) - 1} d\nu$$



The probability of an oscillator occupying a particular energy level

$$P = \frac{\exp\left(-\frac{h\nu}{kT}\right)}{\sum \exp\left(-\frac{h\nu}{kT}\right)}$$



$$P = \frac{\exp\left(-\frac{h\nu}{kT}\right)}{1 + \exp\left(-\frac{h\nu}{kT}\right) + \exp\left(-\frac{2h\nu}{kT}\right) + \dots}$$

# Energy

$$\begin{aligned}\overline{E_\nu} &= \sum_{n=0}^{\infty} E_n p(n) = \frac{\sum_{n=0}^{\infty} E_n \exp(-E_n/kT)}{\sum_{n=0}^{\infty} \exp(-E_n/kT)} \\ &= \frac{\sum_{n=0}^{\infty} nh\nu \exp(-nh\nu/kT)}{\sum_{n=0}^{\infty} \exp(-nh\nu/kT)}\end{aligned}\quad (10.8)$$

$$\begin{aligned}\overline{E_\nu} &= h\nu \frac{\sum_{n=0}^{\infty} nx^n}{\sum_{n=0}^{\infty} x^n} = h\nu \frac{(x + 2x^2 + 3x^3 + \dots)}{(1 + x + x^2 + \dots)}, \\ &= h\nu x \frac{(1 + 2x + 3x^2 + \dots)}{(1 + x + x^2 + \dots)}.\end{aligned}$$

$$\frac{1}{(1-x)} = 1 + x + x^2 + x^3 + \dots$$

$$\frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \dots$$

$$\overline{E} = \frac{h\nu \, x}{1-x} = \frac{h\nu}{x^{-1}-1} = \frac{h\nu}{\mathrm{e}^{h\nu/kT}-1}.$$

$$d\rho(\lambda, T) = \frac{8\pi < E >}{\lambda^4} d\lambda = \frac{8\pi}{\lambda^4} \frac{h\nu}{\exp\left(\frac{h\nu}{kT}\right) - 1} d\lambda = \frac{8\pi}{\lambda^5} \frac{hc}{\exp\left(\frac{h\nu}{kT}\right) - 1} d\lambda$$

$$\rho_\lambda = \frac{8\pi}{\lambda^5} \frac{hc}{\exp\left(\frac{h\nu}{kT}\right) - 1} = \frac{c_1 \lambda^{-5}}{\exp\left(\frac{C_2}{\lambda T}\right) - 1}$$

$$\frac{d\rho_\lambda}{d\lambda} = 0$$

$$\lambda_{max} * T = constant$$

# Stefan's Law

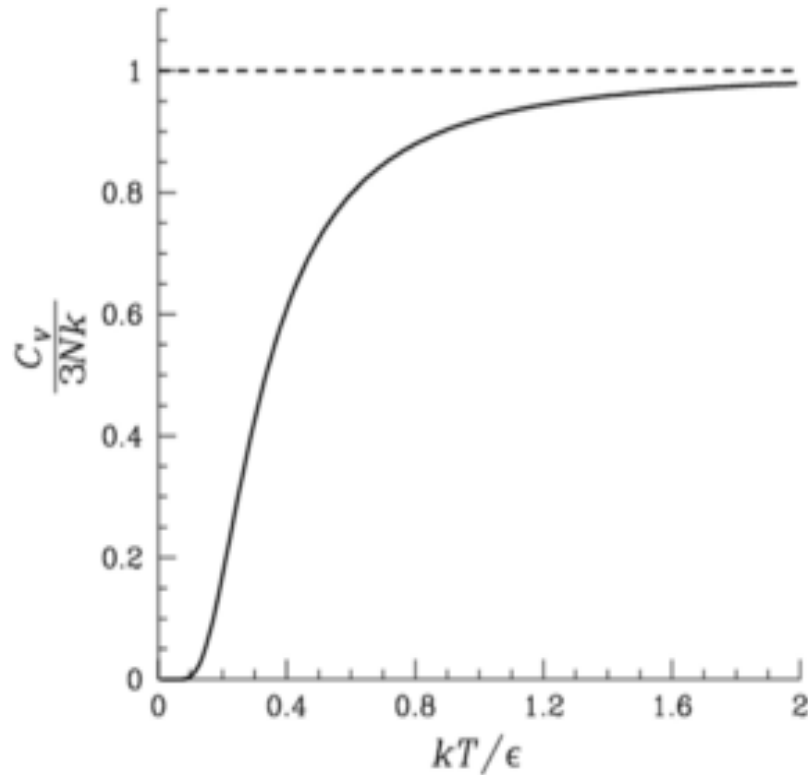
$$E = \int_0^{\infty} d\rho(\nu, T) = \int_0^{\infty} \frac{8\pi h}{c^3} \frac{\nu^3}{\exp\left(\frac{h\nu}{kT}\right) - 1} d\nu$$

$$E = \sigma T^4$$



- The intensity (or flux) at all wavelengths increases as the temperature of the blackbody increases.
- The total energy being radiated (the area under the curve) increases rapidly as the temperature increases (Stefan–Boltzmann Law).
- Although the Rayleigh-Jeans law works for low frequencies, it diverges at high  $\nu$
- This divergence for high frequencies is called the **ultraviolet catastrophe**.

# Heat capacities of monoatomic solids



➤ Classical physics predicts a constant value ( $25 \text{ JK}^{-1}\text{mol}^{-1}$ ) for the molar heat capacity of monoatomic solids.

➤ Experiments at low temperatures, however, revealed that the molar heat capacity approaches zero when temperature approaches zero.

➤ Assumption of discrete energy levels (a collection of harmonic oscillators) again led to a model that matched the experimental observations (Einstein (1905)).

$$C_V = \left( \frac{\partial E}{\partial T} \right)_V = 3Nk \frac{\left( \frac{\theta_E}{T} \right)^2 \exp \frac{\theta_E}{T}}{\left( \exp \frac{\theta_E}{T} - 1 \right)^2}$$

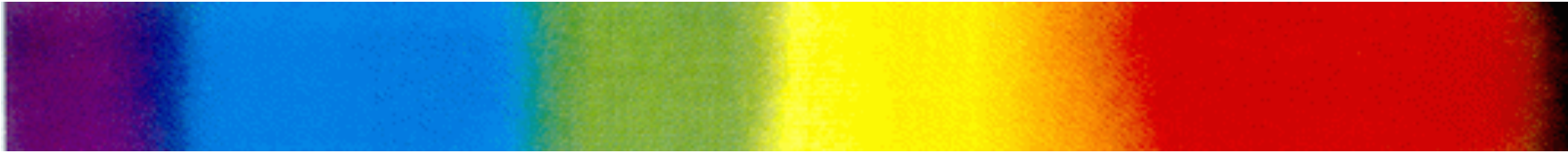
# Continuous Spectra

**Radiation is distributed over all frequencies, not just a few specific frequency ranges.**

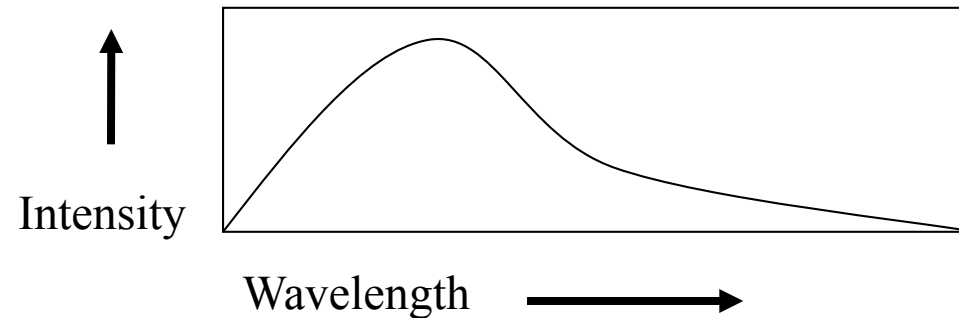


# Continuous Spectrum

- Any hot body produces a continuous spectrum
  - if it's hot enough it looks something like this

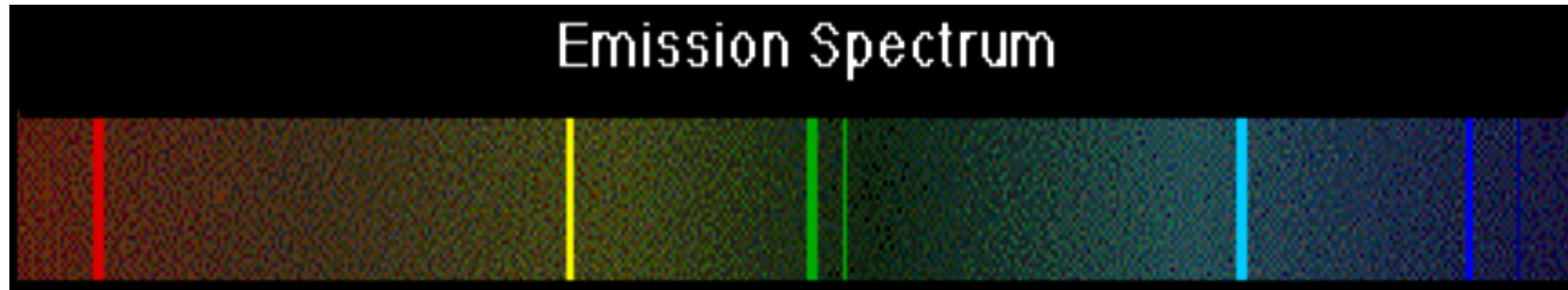


- digitally like this



# Line Spectra

**Pattern of bright spectral lines  
produced by an element.**

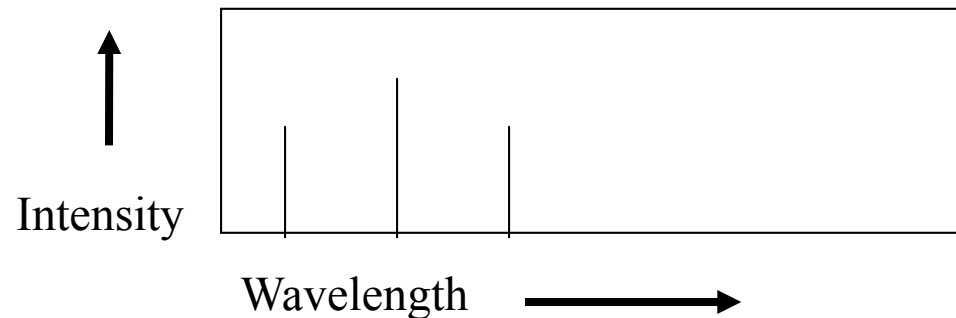


# Line Spectra

- Any gas to which energy is applied, either as heat or a high voltage, will produce an emission line spectrum like this

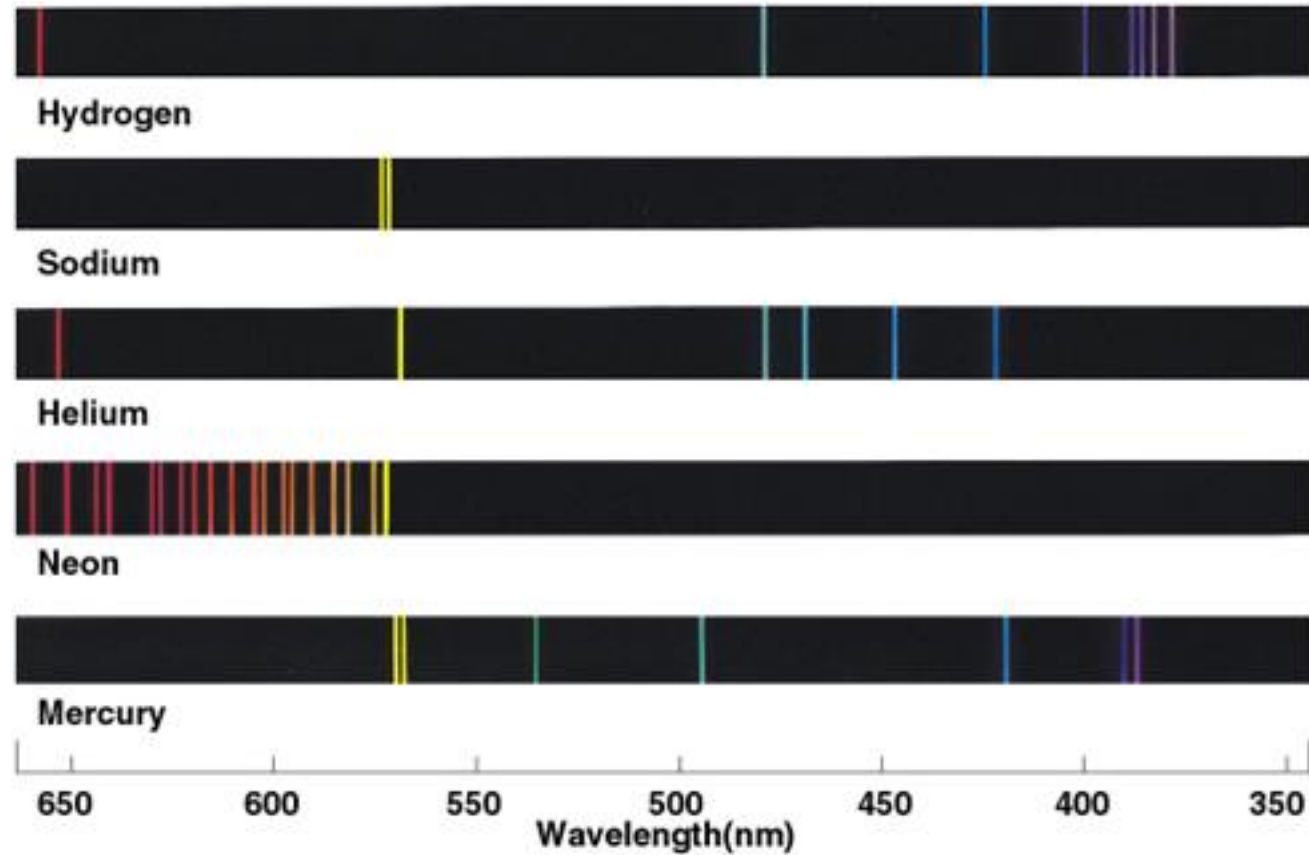


- or digitally like this



# Emission Lines

- Every element has a DIFFERENT finger print.



# Absorption Spectroscopy

**Pattern of dark spectral lines  
where light within a number of  
narrow frequency ranges has been  
removed.**

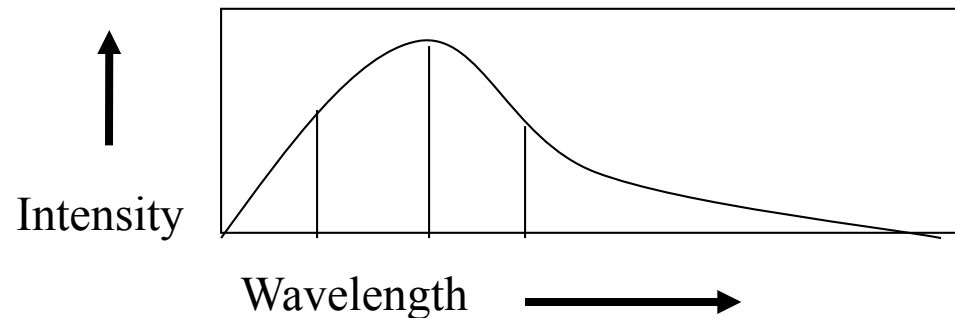




- Any gas placed between a continuous spectrum source and the observer will produce a absorption line spectrum like this

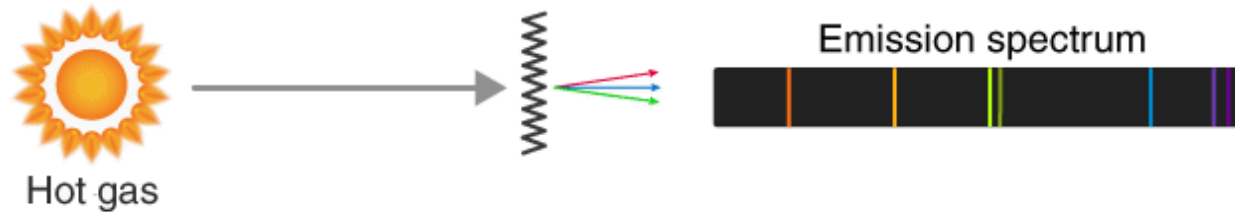


- or digitally like this

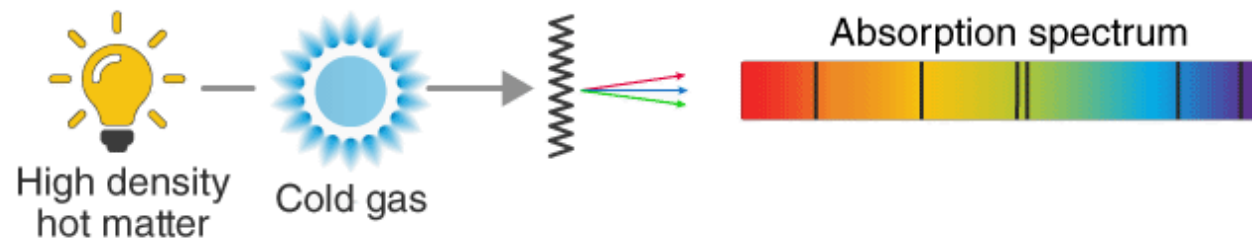


- Atomic and molecular spectra: Absorption and emission of electromagnetic radiation (i.e., photons) by atoms and molecules occur only at discrete energy values.
- Classical physics would predict absorption or emission at all energies.

**(a) Emission Spectra**



**(b) Absorption Spectra**



**Transition between two energy levels and associated wavelength:**

On absorption of light of energy  $\Delta E$ , the species go to higher energy level.

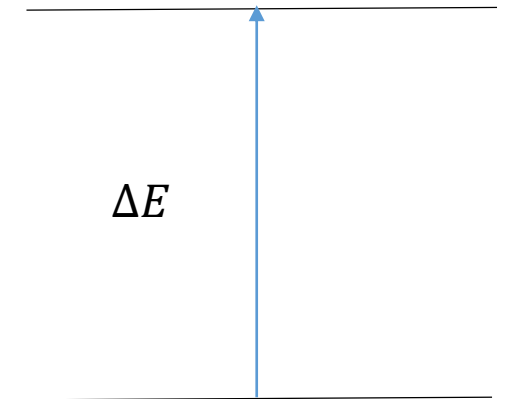
$$\Delta E = h\nu = \frac{hc}{\lambda}$$

$h$ = Planck's constant

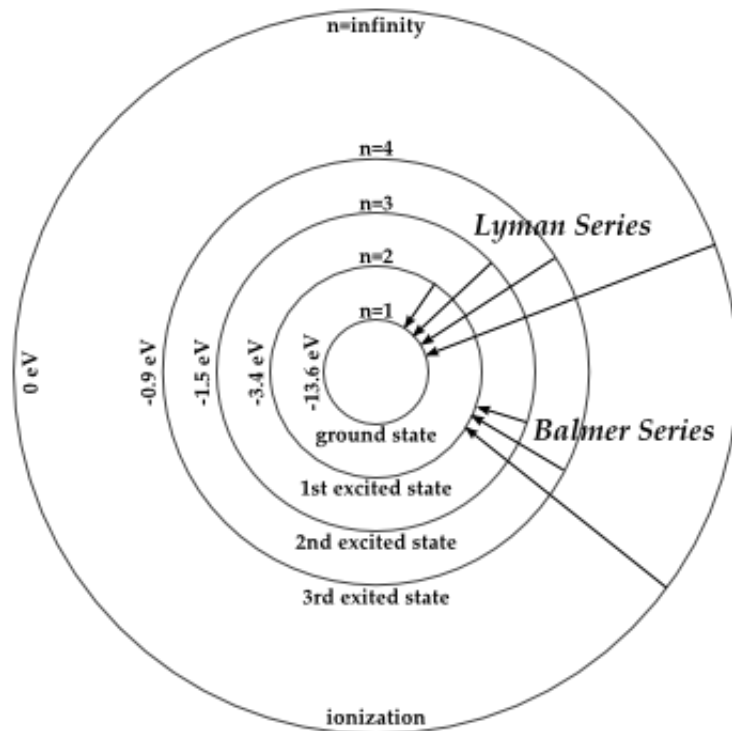
$\nu$ =frequency

$c$ = velocity of light

$\lambda$ =wavelength of light



# Quantization of Energy

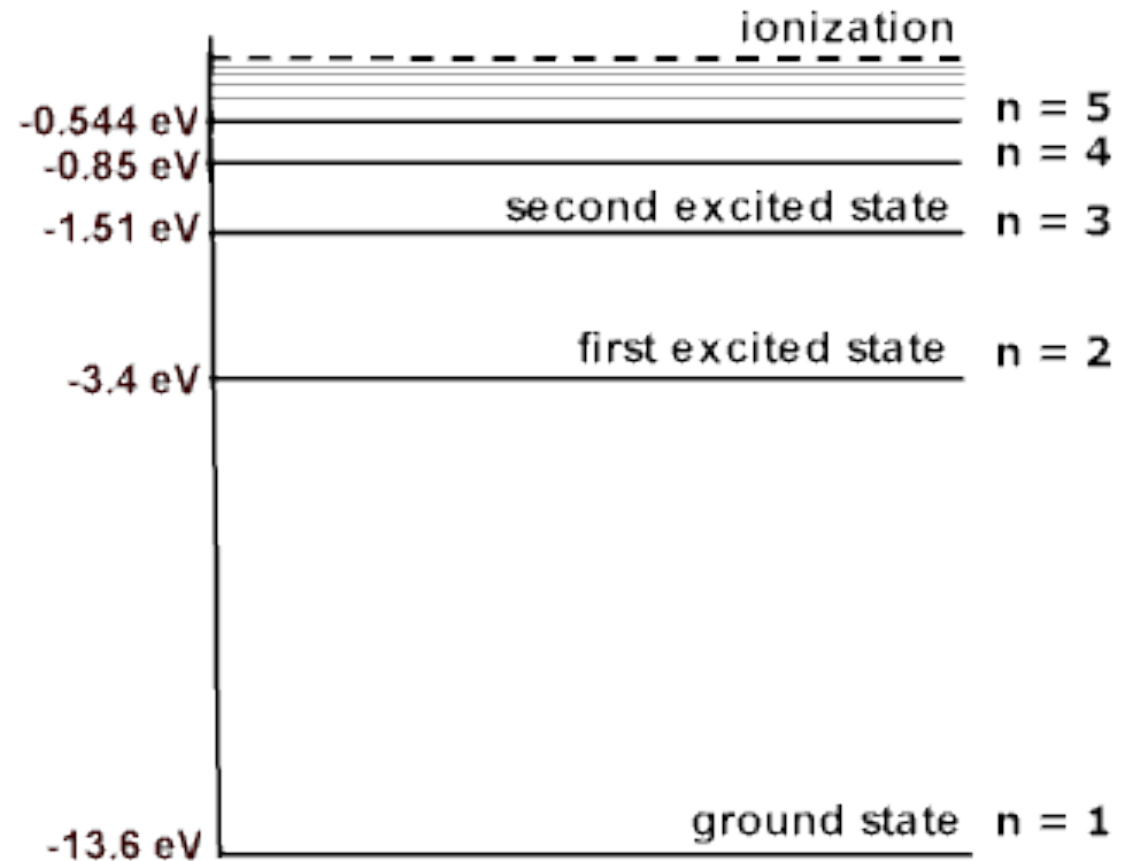


Hydrogen atom

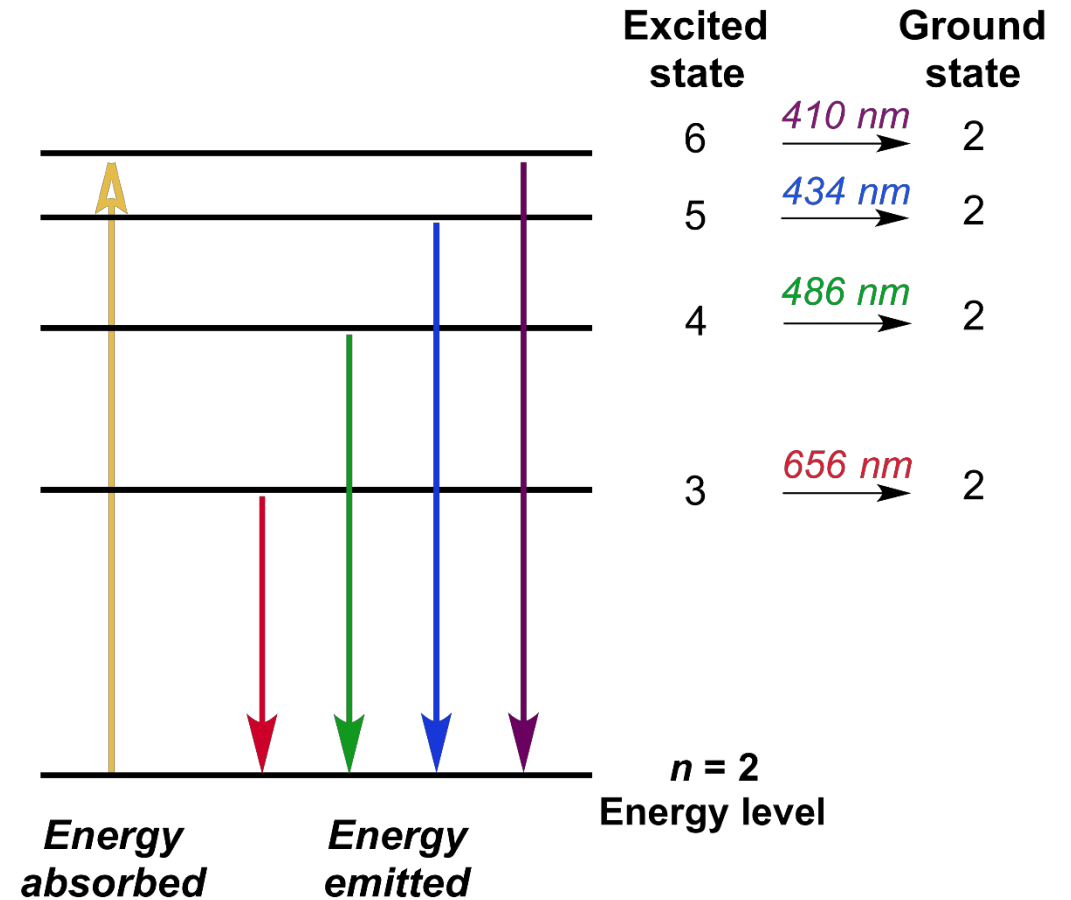
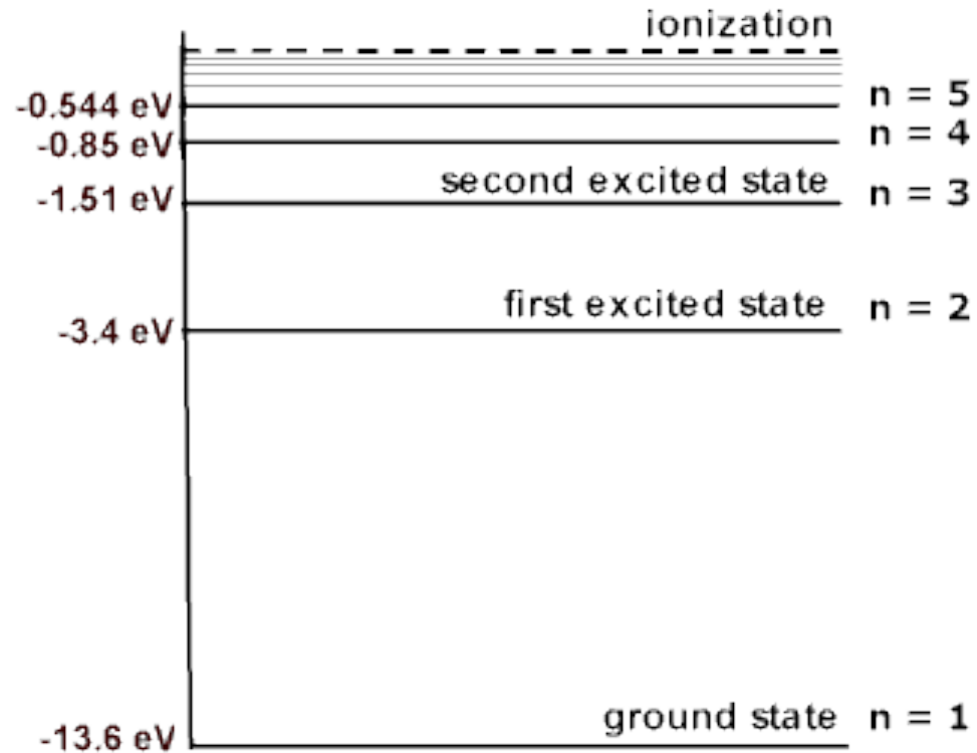
- Electrons can not have all value of energy.
- Electron can only have certain value of energy.
- In hydrogen atom, Electron only exist with energy -13.6 eV, -3.4 eV, -1.5 eV.....
- Electrons with energy -13.6 eV is said to be in a ground state.....
- Electrons can not exist between these energy levels.

# Quantization

- Light can give all the energy to electron or it gave none of the energy to electron
- An electron in the GS can absorb light of 10.2 eV, 12.01 eV, 12.75 eV.....
- Thus, if a light of 10.2 eV struck the atom, the electron will jump from ground state to first excited state.



# Quantization of Energy



<https://www.khanacademy.org/science/physics/quantum-physics/atoms-and-electrons/a/bohrrs-model-of-hydrogen>